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Abstract

die Zusammenfassung

Preface

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1 Introduction

1.1 Motivation

1.2 Tasks and Requirements

The scope of this work is categorized into four primary technical outputs: Robot Setup and Scenario Execution, Dataset Creation, Ground Truth Labeling, and Benchmark Criteria and Evaluation.

1.2.1 Robot Setup and Scenario Execution

This phase focuses on delivering fully operational platforms and execution scripts. It requires the successful software bring-up and deployment of the target mobile robots within the test environment. A precise characterization of each robot's embodiment is essential, capturing physical parameters (such as mass, geometry, footprint, and center of mass), kinematic limits (maximum velocity, acceleration, and braking distance), and sensor payload configurations. To ensure consistent replay across multiple runs, deterministic trajectory execution routines must be implemented. Safe operation is guaranteed through the integration of hardware and software override protocols, including e-stops and speed governors, to prevent collisions during dynamic testing. Finally, scenarios are parameterized and structured into families that systematically vary environmental and dynamic factors.

1.2.2 Dataset Creation

The primary deliverables for this stage are the raw and processed dynamic recordings. This entails acquiring a comprehensive multimodal dataset that spans all execution runs and parameter variations to ensure complete scenario coverage. Sensor streams—including LiDAR, RGB-D, IMU, and wheel odometry—must be standardized into well-documented ROS bag or MCAP formats, maintaining precise temporal synchronization and a structured data schema throughout the dataset.

1.2.3 Ground Truth Labeling

This output yields the annotated dataset records. It involves the development of a systematic ground-truth labeling pipeline to annotate high-precision spatial positioning, bounding box volumes, dynamic trajectories, and state classifications across all recorded datasets.

1.2.4 Benchmark Criteria and Evaluation

The final phase delivers the evaluation framework and associated metrics. This requires formulating standardized quantitative metrics for risk scoring to assess embodiment-dependent safety, maneuver success, and risk thresholds. These metrics will enable cross-embodiment comparative analysis across differing robotic platforms and environmental configurations.

1.3 Contributions

1.4 Outline of the Report

2 Literature Review

3 Conception

This chapter outlines the theoretical and experimental framework developed to evaluate embodiment-dependent risk assessment. Translating the requirements established in Chapter 1 into a concrete methodology, this chapter defines the core scenario families, characterizes the selected robot embodiments, and establishes the ground-truth annotation strategy. Furthermore, it details the formulation of the benchmark criteria necessary to quantify how effectively risk assessment models adapt to changes in a robot's physical configuration.

3.1 Scenarios

Based on our literature review, it became evident that mobile robots are not inherently engineered to detect embodiment-specific hazards, relying heavily on artificial intelligence and vision-language models to identify and assess these risks. To address this gap and evaluate how physical and kinematic variations across mobile robot embodiments influence operational risk, we introduce a structured methodology centered on generating datasets of controlled scenario families. By systematically altering environmental parameters, obstacle behaviors, and platform configurations, we can isolate and quantify embodiment-dependent risk metrics in autonomous mobile robot operations.

We shortlisted the following three distinct scenario categories to form the core of our controlled dataset, each designed to evaluate operational hazards across different contexts without requiring direct physical human-robot interaction:

3.1.1 Blind Corner Navigation

This scenario models a classic high-risk indoor navigation challenge where a mobile robot navigates around an obstructed corner while a human pedestrian simultaneously approaches the same junction. The operational risk in this scenario is driven by two systematically varied factors:

- **Robot Embodiment:** Variations in kinematic limits, mass, stopping distance, and physical footprint drastically alter the hazard profile. Heavier or faster embodiments increase collision severity and require larger safe stopping margins, while larger footprints reduce the available clearing distance around corners.
- **Presence of a Person:** The speed, trajectory offset, and visibility of a pedestrian dictate the likelihood of an encounter. Obstructed sightlines hide incoming pedestrians, reducing the dynamic reaction time for both the robot and the human, thereby substantially elevating localized risk.

3.1.2 Adverse Surface Transitions

This scenario evaluates stability, traction, and locomotion reliability when the robot traverses challenging or inconsistent floor conditions. Examples include navigating over industrial floor grilles, uneven thresholds, or liquid spills. The associated risk profile changes drastically depending on the robot's locomotion paradigm (e.g., wheeled versus legged) and center of mass, which dictate its susceptibility to slipping, tripping, or falling.

3.1.3 Dynamic Social Proximity

This scenario focuses on spatial navigation through environments occupied by inattentive or socially engaged pedestrians. Examples include the robot passing behind an individual distracted by a phone call, or safely maneuvering around a static group of conversing people. The risk metrics evaluate path prediction and proximity maintenance, where the robot's overall size, momentum, and operational volume dictate the safety margins required to navigate the space without startling the pedestrians.

4 Implementation

5 Evaluation

6 Discussion and Limitations

7 Conclusion and Future Work

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